AI-Enhanced Methodologies and Agents for Business Analysis in Software Development

Adaptive Prompt Engineering, Ethical Prompting, and Hybrid AI Agents for Business Analysis

Research Proposal
CSC4046 – Individual Research Project

Submitted by:
A.P.N.K. Abeysekara
SC/2021/12375

Supervisor:
Dr. S.W.A.M. Upamalika

Bachelor of Computer Science (Hons)
University of Ruhuna
February 23, 2026

Declaration

I hereby declare that this research will be conducted by me and will be my own effort, with no part plagiarized and with proper citations. This research proposal is submitted in partial fulfillment of the requirements for the course unit **CSC4046 — Individual Research Project**, under the supervision of **Dr. S.W.A.M. Upamalika**.

Student Details:

- **A.P.N.K. Abeysekara**
  Student ID: SC/2021/12375
  E-mail: `nimnadakavishwarank.ugc@gmail.com`

Supervisor Details:
Dr. S.W.A.M. Upamalika
B.Sc. (Colombo, S.L.), PhD (Ruhuna, S.L.)
Department of Computer Science
University of Ruhuna
E-mail: `samarawikcramau@dcs.ruh.ac.lk`

Signature: ____________________

Date: 23rd February 2026

Contents

1 Introduction **1**

1.1 Introduction 1

1.2 Problem Definition 1

1.3 Objective of the Research 2

2 Literature Review and Related Work **3**

2.1 Preliminary Literature Review 3

2.2 Related Work 3

3 Methodology **5**

3.1 Material 5

3.2 Methods 5

3.3 Time Plan 7

Chapter 1

Introduction

1.1 Introduction

Business Analysts (BAs) serve as critical intermediaries in software development, bridging the gap between client expectations and technical implementation across the Software Development Life Cycle (SDLC). In modern software teams, BAs must translate stakeholder needs into clear requirements, maintain alignment across Agile roles, and reduce delivery risks caused by misunderstandings and ambiguous specifications.

In parallel, Artificial Intelligence (AI) adoption has increased in software engineering. While many AI tools focus on developer productivity (e.g., coding assistants), BA-specific support remains limited—particularly for human-centric tasks such as interpreting stakeholder intent, resolving ambiguity, and documenting requirements with ethical safeguards. In Small and Medium Enterprises (SMEs), resource constraints and limited AI skills can amplify these challenges, making BA work time-consuming and prone to misalignment and rework.

This research is motivated by the need to **augment** BAs with AI (rather than replacing them). The proposed solution is to develop:

- **Adaptive prompt engineering methodologies** that improve requirements elicitation through iterative refinement;
- **Ethical prompting and bias-reduction checks** to improve transparency and trust; and
- **Hybrid AI agents** (human-in-the-loop) that support BA workflows and Agile collaboration.

The work will be designed and evaluated in the context of Sri Lankan SMEs to produce practical, measurable improvements in BA tasks.

1.2 Problem Definition

The core problem is the **underutilization and poor fit** of current AI approaches for BA processes in software development, leading to inefficiencies, inaccuracies, and ethical risks. Specifically:

1. **Ambiguity in requirements:** Elicited requirements often contain vague terms

or missing details. Generic AI use may not ask the right clarification questions, resulting in incomplete or misleading statements.

2. **Ethical risk and low transparency:** AI-generated artifacts can reflect bias in wording or prioritization. Without explicit checks and clear explanations (assumptions/uncertainty), teams may not trust or adopt AI-assisted outputs.
3. **Alignment and handover friction:** In SME Agile teams, limited BA tooling and AI familiarity can widen communication gaps between stakeholders, BAs, and developers, leading to rework and delivery delays.

Research Questions:

Main RQ: How can AI-enhanced methodologies and hybrid agents improve BA work in software development while ensuring ethical safeguards and preserving the BA's human role?

- **SQ1 (Elicitation):** How well do adaptive prompt workflows reduce ambiguity and improve requirement clarity/completeness?
- **SQ2 (Ethics):** Which ethical prompting and bias checks best improve transparency and reduce bias in BA artifacts?
- **SQ3 (Hybrid workflow):** How much does a human-in-the-loop hybrid agent prototype improve BA efficiency and BA–developer alignment in Agile SMEs?

1.3 Objective of the Research

The primary objective is to **develop and evaluate AI-enhanced tools** that improve BA efficiency, accuracy, and ethical practices in software projects.

Secondary objectives:

1. **Adaptive prompt methodologies:** enhance requirements elicitation and refinement; hypothesized **20–40%** improvement on controlled tasks via **A/B comparisons** and rubric-based quality ratings (clarity/completeness).
2. **Ethical prompting and bias checks:** improve transparency and reduce bias; hypothesized **15–30%** reduction in bias-related issues via checklists and simple bias-lint metrics on BA artifacts.
3. **Hybrid AI agents:** support BA workflows and Agile collaboration; hypothesized **30–50%** improvement on selected efficiency/alignment metrics in controlled workflow trials with human-in-the-loop review.
4. **Dataset and insights:** curate an anonymized dataset (where permitted) and produce practical guidance for BA adoption in SMEs.

Expected impact: Reduced project risks, improved BA productivity, upskilled SME teams, and a scalable toolkit aligned with industry best practices (with potential alignment to IIBA perspectives).

Chapter 2

Literature Review and Related Work

2.1 Preliminary Literature Review

A preliminary review of recent sources (2023–2026), including Google Scholar, IEEE Xplore, and arXiv, suggests that while AI is increasingly discussed in software engineering [8], **BA-focused coverage remains relatively limited**. Existing studies often emphasize general productivity gains (e.g., faster documentation, automated summaries) but provide limited support for BA-centered requirements elicitation workflows, stakeholder alignment, and ethical safeguards.

Most studies evaluate outputs on artifacts such as **user stories**, **SRS sections**, **acceptance criteria**, or **meeting summaries [4, 5].**, which are central to BA work but still require careful handling of ambiguity, traceability, and stakeholder fairness.

For example, research on generative AI in Agile software development highlights improvements in automation and velocity, yet commonly lacks [7]:

- BA-tailored prompt methodologies for elicitation and clarification;
- explicit handling of **stakeholder intent, context, and socio-cultural nuances** (including emotions and tacit knowledge) in requirements interpretation; and
- systematic ethical checks to reduce bias and improve trust in generated artifacts [10, 9].

Strengths of prior work include evidence that LLM-based tools can improve documentation speed and assist communication tasks. However, this preliminary scan suggests that BA-specific work addressing SME constraints, sustained **human-in-the-loop refinement**, and BA-oriented evaluation metrics (e.g., requirement clarity and stakeholder fairness) remains limited [3, 14]. These gaps motivate this research for practical adoption in Sri Lankan SMEs (see [2, 1]); LLM-based requirements elicitation is discussed in [**?**].

2.2 Related Work

This proposal builds on three research directions that have recently gained attention in software engineering and human–AI interaction:

- **LLM prompt engineering for structured text tasks:** Prior studies suggest that structured prompts and iterative refinement can improve output quality for complex language tasks, motivating their use for requirements drafting and clarification [8, 13].

- **Agentic workflows for task decomposition:** Multi-agent or role-based workflows split complex work into drafting, critique, and validation steps, suggesting a pathway to support BA activities through modular assistants [6, 15].
- **Trustworthy and ethical AI for text generation:** Work on responsible AI highlights risks such as bias and ungrounded outputs, motivating lightweight transparency and fairness checks for AI-generated artifacts [10, 9].

Novel contribution of this proposal: Rather than studying these directions separately, this research combines them into a single BA-oriented workflow tailored to requirements work in Sri Lankan SMEs. Concretely, the proposed system will:

- implement an **elicitation loop** that generates clarification questions and refines requirements iteratively [3, **?**] (not only drafting text once);
- apply a **BA-document ethics layer** (bias-sensitive phrasing, stakeholder coverage checks, and explicit assumptions/traceability prompts [4]); and
- orchestrate these steps in a **human-in-the-loop agent workflow** that supports BA–developer alignment, evaluated using task-based measures and practitioner feedback.

Chapter 3

Methodology

3.1 Material

The research will use a combination of primary and secondary data:

1. **Custom dataset from Sri Lankan SMEs (target: 15–25 BAs):**
   - Anonymized interview transcripts (requirements challenges, BA workflows, AI readiness)
   - Survey responses on AI usability and perceived impact
   - Sample SDLC artifacts (requirements documents, user stories, change requests) where permitted
   - **Ethics/data handling (initial):** informed consent; de-identification (remove names/organizations); secure storage with restricted access; right to withdraw before analysis/reporting.
2. **Open/benchmark datasets (for baseline comparisons):**
   - Public requirements text samples (e.g., open repositories or curated datasets)
   - Synthetic/simulated requirement sets if access is limited
3. **Tools and resources:**
   - Python-based NLP/LLM tooling (e.g., Hugging Face ecosystem) for experimentation
   - Bias/fairness measurement utilities (e.g., AIF360 or equivalent bias metrics adapted for text [9])
   - Prototype environment (e.g., Streamlit or lightweight web stack) for the BA toolkit

3.2 Methods

A **mixed-methods** design will be used, combining **Design Science Research** (prototype building) with **empirical evaluation**.

Step 1: Design and Development (Design Science)

1. **Adaptive Prompt Methodology:** Develop an iterative prompting approach for requirements elicitation, using:

- initial stakeholder input → draft requirement
    - ambiguity detection → targeted clarification questions
    - refinement loop → improved requirement/user story

2. **Ethical Prompting + Bias Reduction Framework:** Create a practical checklist/DSL-like rubric for BA documents, including:
    - bias-sensitive language detection
    - stakeholder fairness prompts (who is affected/ignored?)
    - transparency notes (assumptions, uncertainty, traceability)

3. **Hybrid AI Agents Prototype (Human-in-the-Loop):** Implement a prototype BA toolkit that orchestrates components such as:
    - Elicitation Assistant (adaptive prompting)
    - Ethics/bias checker (rubric + simple metrics)
    - Documentation generator (user stories/acceptance criteria/SRS snippets)
    - Human feedback loop (BA edits and acceptance decisions)

Step 2: Evaluation (Empirical Validation)

Evaluation will include both quantitative and qualitative measures.

Quantitative evaluation (initial):

- **Efficiency:** time-to-complete BA tasks (baseline vs AI-supported), task throughput
- **Accuracy/quality:** requirement clarity score (rubric-based [4, 5]), defect/ambiguity count reduction
- **Ethics/bias:** checklist compliance and simple bias flags; transparency score

Operational definitions (initial; refined after a pilot):

- **Requirement quality:** 1–5 rubric for clarity, completeness, and testability.
- **Transparency:** presence/absence of assumptions, uncertainty notes, and traceability links to source input.
- **Bias/inclusivity:** inclusive-language flags + lightweight stakeholder-coverage checklist.

Qualitative evaluation (initial):

- BA interviews on usefulness, trust, adoption barriers
- Thematic analysis of feedback (pain points, improvements, ethical concerns)

Experimental setup: Where possible, small A/B-style comparisons will be conducted (manual BA workflow vs AI-augmented workflow) on representative tasks in SME-like scenarios.

3.3 Time Plan

The initial plan spans approximately 8 months and may be revised based on access to participants and early findings.

Table 3.1: Tentative time plan

Month	Major Tasks and Subtasks
1	Detailed literature review; finalize research questions; confirm evaluation metrics and instruments (survey/interview guides).
2	Design adaptive prompting methodology; draft ethical framework; architecture design for hybrid agents; prototype wireframes.
3–4	Data collection: recruit 15–25 BAs/SMEs; conduct interviews and surveys; curate/anonymize artifacts; prepare datasets.
5–6	Prototype implementation (web-based toolkit); integrate prompt engine + ethics checks; pilot testing; iterative refinement.
7–8	Full evaluation (quant + qual); analysis and interpretation; write thesis/proposal updates; prepare paper/report for submission.

Risks and mitigation (initial):

- **Limited data access:** use public/simulated artifacts and keep the prototype evaluation task-based.
- **Low participation/availability:** recruit via professional networks and schedule short, structured sessions.
- **Model limitations or unsafe outputs:** keep a human-in-the-loop review, log assumptions/uncertainty, and avoid sensitive or identifying data.
- **Tool adoption risk:** iterate early with pilot feedback and keep the workflow lightweight for SME teams.

Bibliography

[1] Deloitte. (2026). *Tech Trends 2026*. https://www.deloitte.com/us/en/insights/topics/technology-management/tech-trends.html

[2] IIBA. (2024). *Futureproof: Amplifying Agility with AI and Insightful Business Analysis*. IIBA Publications.

[3] IIBA. (2015). *A Guide to the Business Analysis Body of Knowledge (BABOK Guide) (Version 3)*. International Institute of Business Analysis.

[4] ISO/IEC/IEEE. (2018). *ISO/IEC/IEEE 29148:2018 — Systems and software engineering — Life cycle processes — Requirements engineering*.

[5] Wiegers, K. E., & Beatty, J. (2013). *Software Requirements* (3rd ed.). Microsoft Press.

[6] Wooldridge, M. (2009). *An Introduction to MultiAgent Systems* (2nd ed.). Wiley.

[7] International Journal of Scientific Research in Artificial Intelligence and Machine Learning. (2024). *Integrating Generative AI for Advancing Agile Software Development and Mitigating Project Management Challenges*. https://thesai.org/Downloads/Volume15No3/Paper_6-Integrating_Generative_AI_for_Advancing_Agile_Software.pdf

[8] DFKI. (2025). *Generative AI in Software Engineering: Transforming the Software Development Process*. https://www.dfki.de/fileadmin/user_upload/DFKI/Medien/News/2025/Wissenschaftliche_Exzellenz/Generative_AI_in_Software_Engineering_Transforming_the_Software_Development_Process_2025.pdf

[9] Bellamy, R. K. E., et al. (2019). *AI Fairness 360: An Extensible Toolkit for Detecting and Mitigating Algorithmic Bias*. IBM Journal of Research and Development.

[10] Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). *On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?* (Conference paper).

Skimmed papers (initial reading list).

[11] *Hybrid Methodologies in Software Development with AI: The Role of Business Analysis in Economic Optimization of Personalized Medicine*. (Nov. 2025). SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5579173

[12] *AI-driven business analytics and decision making*. (Aug. 2024). WJARR. https://wjarr.com/sites/default/files/WJARR-2024-3093.pdf

[13] *Integrating Generative AI into the Software Development Lifecycle: Impacts on Code Quality and Maintenance*. (Aug. 2024). IJSRA. https://ijsra.net/sites/default/files/IJSRA-2024-1837.pdf

[14] ITWorx. (2025). *The Business Analyst Guide (AI and Problems)*. https://www.itworx.com/wp-content/uploads/2025/03/The-Worxian-Wizard-Software-Business-Analysis-1.pdf

[15] *Artificial Intelligence Applications in Lean Startup Methodology: A Bibliometric Analysis of Research Trends and Future Directions*. (Jun. 2025). arXiv. https://www.arxiv.org/pdf/2512.22164

[16] *AI-Driven Business Intelligence: Unlocking the Future of Decision-Making*. (Sep. 2023). ESP Journals. https://www.espjournals.org/IJACT/2023/Volume1-Issue2/IJACT-V1I2P108.pdf

[17] *Integrating AI to Enhance Business Analysis Education*. (Dec. 2024). KoreaScience. https://koreascience.kr/article/JAKO202405656880175.page

[18] *AI-enhanced Business Intelligence for decision-making*. (2025). ScienceDirect. https://www.sciencedirect.com/science/article/pii/S1877050925028303